

Gautam Galada

Ph.D Student - Researcher

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[LinkedIn](#) • [Github](#) • [Google Scholar](#)

Objective

Innovative computer science graduate with extensive interest and practical knowledge in Natural Language Processing and Pattern Recognition, seeking to obtain an opportunity with focus on core neural network functionalities. I practice making latent spaces interpretable and optimizing them through internal and external understanding. Also, I ask questions. A lot of questions. Questions with answers and questions without answers, and then I try to find the answers to the questions that don't have one. Currently I am working on designing probing strategies for Explainable AI (XAI).

Experience

AI&ML Intern

Buffalo Niagara Medical Campus, Inc.

Mar 2025 - July 2025

- Built CV systems for anomaly detection and feature extraction, implementing physics-informed ML models (SINDy - SHRED) for fluid dynamics using sparse neural networks.
- Engineered MCP framework integrations with prompt engineering and evaluation protocols, delivering scalable AI agents and knowledge synthesis tools. Led HIPAA-compliant ML development for secure medical data processing, optimizing models and advancing privacy research through automated workflows and data masking techniques.

Co-Founder

Digital Fortress Pvt. Ltd.

Oct 2021 - Sept 2023

- Founded and led a passwordless authentication startup, developing FIDO2-compliant biometric solutions with advanced facial recognition and liveness detection. Established zero-knowledge proofs and decentralized identity protocols, achieving a 99.9% reduction in account takeovers and a 40% increase in user adoption rates compared to traditional 2FA methods

Research Assistant

Artificial Intelligence and Robotics Center, VIT-AP

Oct 2021 - May 2023

- Developed advanced lip synchronization modules using VQGAN and Codebook algorithms, improving mouth movement accuracy by 40% and reducing visual artifacts by 60%. Achieved a lip-sync error rate below 100 milliseconds, resulting in highly realistic talking face animations.
- Led a project on federated learning for cyberbullying detection using zero-shot learning and a CNN-BiLSTM-Attention mechanism, integrated with custom GloVe and context vector analysis. The system achieved an accuracy of 97.4% for English and 92.7% for Hindi, demonstrating effectiveness across multiple languages

AI Product Developer

RAPYD AI Gmbh

Jan 2021 - June 2021

- Enhanced abstractive summarization of podcast content by refining transformer and PEGASUS models, achieving a 35.69% improvement in ROUGE-L score, reducing factual inconsistencies by 25%, and increasing summary coherence by 30%.
- Conducted in-depth statistical and linguistic analysis of Spotify's podcast corpus, deriving insights to augment model precision and facilitate smoother integration into user-focused applications.

Education

University of North Texas

Ph.D in Computer Science and Engineering - GPA (4.0)
Smart Electronics System Lab

Denton, TX

Aug 2025 - Present

University at Buffalo

Master of Science in Artificial Intelligence - GPA (3.12)

Buffalo, NY

Aug 2023 - Dec 2024

VIT-AP UNIVERSITY

B. Tech in Computer Science and Engineering with specialization in AI (CGPA: 8.51)

Andhra Pradesh, India

July 2019 - May 2023

Publications

- Gautam Galada, Dev Kapadia, Sibi Chakkaravarthy Sethuraman, Sunil Kumar Singh, Anupama Namburu, and Hari Seetha. 2018. DUCK: Developing Unassisted Cognitive Knowledge (DUCK) for Cyberbullying detection . In . ACM, New York, NY, USA, 26 pages. ([In-Press](#))
- Galada, Gautam & Chakkaravarthy, Sibi. (2024). Optimizing Policy Gradient Methods for Adaptive Gait Rehabilitation. 10.13140/RG.2.2.25370.04807/1.
- Sibi Chakkaravarthy Sethuraman, Aditya Mitra, Gautam Galada, Anisha Ghosh, and S Anitha, Metakey: A Novel and Seamless Pass- wordless Multifactor Authentication for Metaverse, in 2022 IEEE International Symposium on Smart Electronic Systems (iSES), pp. 662–664, 2022.
- Sibi Chakkaravarthy Sethuraman, Aditya Mitra, Anisha Ghosh, Gautam Galada, and Anitha Subramanian. *MetaSecure: A Passwordless Authentication for the Metaverse*. arXiv preprint arXiv:2301.01770, 2023. Available at: <https://arxiv.org/abs/2301.01770>
- Sibi C. Sethuraman, Gaurav R. Tadkapally, Saraju P. Mohanty, Gautam Galada, and Anitha Subramanian. *MagicEye: An Intelligent Wearable Towards Independent Living of Visually Impaired*. arXiv preprint arXiv:2303.13863, 2023. Available at: <https://arxiv.org/abs/2303.13863>

Personal Achievements

- Contributed during the Metaverse demonstration at the "Recent Research Trends and Applications of AI and IoT" FDP (Faculty Development Program) at NIT Warangal.
- Hosted Mindhack, a student-run AI summit where AI pioneers such as Thomas Wolf (HuggingFace CTO), Sam Lightstone (IBM CTO) and other twenty dignitaries provided insights on “Can Machines Think?”
- Winner at Smart India Hackathon 2022
- Top 3 startups at FIDO Developer Challenge India 2022
- Runner-up at Fintech Innovative Challenge 2022

Skills & Interests

Technical: Python, JAVA, Bash, R, C++, SQL, Matlab, Cuda, C#

Frameworks/Tools : Tensorflow, Pytorch, Lightning, Keras, scikit-learn, Numpy, Pandas, open-cv, NLTK, HuggingFace, Langchain, OpenAI, Ollama

Soft Skills: Research and Planning, Communications, Problem Solving, Innovation.

Interests: Open Source, Internet of Things, Quantum Computing, Memory based learning

Research Interests

- Latent Space Interpretability
- Language Modelling
- Natural Language Processing
- Scaling and Optimization Models
- Reinforcement Learning
- Edge AI and Embedded Intelligence
- Physics Informed Machine Learning
- Systems and Philosophy
- Quantum Machine Learning

Best Projects, Hackathons and Competitions

LLM Interpretability: Refusal Mechanisms in Privacy-Sensitive Contexts

- Developing interpretability research examining neural network behavior in privacy-sensitive contexts by training models on synthetic database mappings and analyzing refusal mechanism effectiveness through a dual-phase fine-tuning methodology consisting of a database learning phase followed by refusal training to study the emergence of safety behaviors in language models.

Human-in-the-Loop Prompt Auditing for LLMs (A PROJECT WITH GANSON STREET)

- Building human-in-the-loop audit trail framework on frontier LLM, implementing multi-layer quality control pipelines with custom rubrics. Enhancing domain-specific knowledge through SME collaboration to ensure transparency, compliance, and reliability in production AI systems.

Dynamic Online Neural Network Assistant

- Designing DONNA, an automation framework for providing assistance to researchers. It enables assisted model training, on-fly dataloader and inferencing features, it also enables hardware monitoring and file management along with hyperparameter tuning. An inspiration from “SUITS” television show where every Harvey needs a DONNA.

FAISS Optimized Information Retrieval System

- End-to-end information retrieval system using FAISS-based semantic search and Llama 3 for conversational response generation.
- Leverages dense vector embeddings from domain-specific Wikipedia data (~50,000+ articles), with Maximum Marginal Relevance (MMR) and semantic clustering to optimize relevance-diversity trade-offs during retrieval.
- Integrates query-type inference and context-aware reasoning within a Flask interface, achieving near real-time response latency via cuda accelerated retrieval and transformer-based generation.

Variational Auto-encoder with Chaotic Mapping for Scene Generation

- Variational Autoencoder (VAE) framework enhanced with chaotic systems (Logistic, Henon, and Lorenz maps) in the latent space.
- Aims to mitigate alignment inconsistencies and improve spatial coherence in generated scenes.
- Demonstrates a 15% improvement in diversity metrics over conventional VAE baselines.
- Achieves superior structural and semantic consistency in scene synthesis.
- Exploring chaotic dynamics in enhancing latent space representation, critical to hyper-parameters.

Document Serialization and information retrieval using RAG

- Developed an advanced recruitment processing pipeline using SOTA embedding and chunking strategies for constructing vector and graph knowledge databases with ChromaDB and Neo4J. Enhanced retrieval performance by integrating Llama2 (7b) through knowledge distillation, achieving an F1 score of 83.12.
- Optimized model accuracy and efficiency with LoRA and QLoRA. The evaluation metric resulted in achieving ROUGE (L) score of 32.21; BLEU score of 8.13, ensuring reliable and accurate information retrieval.

Optimizing Policy Gradient methods for Gait Rehabilitation

- Analyzed human gait data using a CNN integrated with Proximal Policy Optimization(PPO) and the Artificial Algae Algorithm through sequential frame NAC value assessment.
- This approach achieved a 99% convergence rate in gait rehabilitation modeling within 50 episodes, highlighting its efficiency and accuracy.
- Additionally, SHAP analysis was applied to enhance model interpretability, ensuring transparent and personalized rehabilitation outcomes.

FIDO Developer Challenge India, 2022 – Top 3 [Certificate](#), INEX 2022 – Silver Award [Certificate](#)

MAuthN

- Implemented facial based anti-spoofing strategies for low end devices.
- Interpretable facial features, enhancement and mapping, pixel acceleration and tracing (Lucas-Kanade Method)
- Related Skills: Flask, Pytorch , Python, Backend, Cloud computing, Linux, IoT

Smart India Hackathon, 2022 – Winner

MediaRF – Transmission of multimedia files securely over police radios

- Encoding files to be transmitted over walkie-talkies / any audio communication device.
- Implemented robust compression algorithms for multimedia files, encryption of the files and encoding using various PSK, FSK and QAM algorithms.
- Related Skills: Cryptography, Python, Modulation-demodulation, file compression.

Hobbies and Interests

Lex Fridman, Huberman Labs (Podcasts); Quantum Computing Since Democritus, Surely You're Joking, Mr. Feynman! (Books); European Football.